1. Methodology

Measuring and predicting financial risk from shipment cost increases caused by delays and disruption events using ERP-style historical data

We are crafting the methodology for using historical ERP-style shipment data to identify and predict the financial risk associated with increased shipment costs arising from delay days and external disruption regimes in logistics and supply chain systems.[1]

3.1.1 Approaches to find the impact of delay

Now, we will be taking you through to steps and our approach to find the impact of delays on shipment cost then on the financial of the organization.

Approaches

(i): Identify the rate and effect of shipment delays caused by external factors on transport cost and procurement performance.

- Measure delay rate and delay severity overall and by disruption type.
- Quantify cost uplift under delay and disruption regimes via correlation and regression.[2,3]

(ii): Forecast shipment delays and estimate the financial impact in the next four months.

- Convert transaction records into a weekly/monthly risk time series.
- Benchmark forecasting models (ARIMA, SARIMA, SARIMAX, Prophet) using rolling-origin backtesting.[4,5]
- Forecast delay incidence (count process) using Poisson and Negative Binomial and translate delay forecasts into expected cost uplift.[6,7]

(iii): Translate ERP-driven predictive insights into mitigation strategies to reduce financial exposure.

- Define risk thresholds and action rules (expedite, reroute, supplier escalation, buffer).
- Connect drivers (delay intensity, disruption mix) to mitigations recommended in SCRM and resilience literature.[2,8,9]

This design deliberately mirrors how procurement leaders govern risk:

measure → forecast → act.[2,3]

3.2 Data source, ERP mapping, and scope

3.2.1 Dataset used (uploaded ERP proxy)

The empirical analysis uses the uploaded shipment dataset as an ERP-like extract. Each record contains:

- Shipping_Cost_USD (financial outcome),
- Order_Weight_Kg (shipment scale),
- Delay_Days (delay magnitude), and
- Disruption_Event (external regime label).[1]

This dataset is treated as a realistic proxy for data typically available from ERP + logistics reporting (ERP purchase-to-pay enriched by shipment and freight settlement fields). The dataset supports robust analysis of cost drivers and regime effects, and it provides the minimum variables needed to define delay-driven cost risk and its uplift.

3.2.2 ERP requirement for true time-series forecasting

To forecast "next four months," the model needs a time anchor (weekly or monthly). In ERP implementations, this is normally one of:

- Goods receipt posting date (GR posting)
- Shipment arrival/ETA milestone date (TMS/LE)
- Freight invoice posting date (FI/CO)

If the uploaded dataset does not include an explicit date column, the forecasting section remains methodologically valid but requires an ERP extract that adds a timestamp. This is consistent with ERP analytics practice and process/event-log literature: operational processes must be time-indexed to support monitoring and prediction.[10]

3.3 Risk construct and variable definitions

3.3.1 Cost - Base cost and Risk uplift

Shipment costs are naturally driven by scale (e.g., weight and volume). If risk is measured using raw cost, the signal is dominated by volume rather than disruption exposure. To make the measure actionable for procurement and finance, this methodology separates:

- a structural base cost driven by scale, and
- a risk uplift attributable to delays and disruption regimes.[2,3]

This approach is aligned with the SCRM principle that risk identification should isolate the effect of shocks and variability rather than conflating them

with normal operating scale.[2,3]

3.3.2 Core variables (shipment-level)

Let each shipment be indexed by i:

- C_i: shipping cost (USD)
- W_i: weight (kg)
- D_i: delay days (count, non-negative)
- Z_i: disruption regime (categorical; converted to dummy variables)

3.3.3 Baseline cost model

Cost data is typically heavy-tailed; a log transform stabilizes variance and makes multiplicative cost relationships more linear. The baseline is estimated using:

$$\log(C_i) = \alpha + \beta \log(W_i) + \varepsilon_i$$

The baseline prediction:

$$\hat{C}_i^{base} = \exp(\hat{\alpha} + \hat{\beta} \log(W_i))$$

3.3.4 Financial risk metric: uplift

Uplift is the percentage cost above what shipment scale alone would predict. In finance terms, $U_i$is a "variance-like" risk measure that can be aggregated over time and compared across regimes**.**

$$\boldsymbol{U_i = \frac{C_i}{\hat{C}_i^{base}} - 1}$$

3.4 Data preparation and statistical controls

3.4.1 Cleaning rules

- Disruption labels: Missing disruption values are treated as "No Disruption" for comparability unless a business rule requires "Unknown."[1]
- Outliers: Retained (because extreme costs are part of financial risk) but modeled using log transforms and robust inference.
- Categorical encoding: Disruption_Event is modeled as a categorical factor with "No Disruption" as the reference category.

3.4.2 Robust standard errors

Freight costs often exhibit heteroskedasticity (variance increases with scale and disruption). Therefore, regression models use heteroskedasticity-consistent (White/HC) standard errors to avoid overstating statistical significance.[11]

3.5 Approach (i): delay rate + correlation + cost impact estimation

3.5.1 Delay rate and severity (risk profiling)

To measure the "rate and effect" of delays, the thesis reports:

- Delay rate: $\Pr(D_i > 0)$overall and by disruption regime
- Delay severity: mean/median delay days overall and by regime
- Cost uplift severity: mean/median uplift and extreme uplift rate by regime

This aligns with SCRM practice: risk is not only magnitude but also frequency, and disruption regimes differ in severity.[2,3]

3.5.2 Correlation analysis (screening relationships)

The thesis estimates:

- We are evaluating Pearson correlation between cost of shipment (. C_i) and all other variables given in dataset.
- Spearman correlation as robustness to heavy tails and nonlinear monotonic patterns for the given dataset.

3.5.3 Regime comparisons (disruption types)

The thesis compares cost distributions across disruption regimes using:

- mean/median costs and uplifts by regime
- ANOVA on $\log(C_i)$where appropriate, and/or Kruskal–Wallis when distributional assumptions are weak

This is theoretically consistent with resilience research: disruptions are not equivalent; their consequences differ by type and system context.[8,9]

3.5.4 Explanatory cost model (marginal effect of delay and disruption on cost)

To quantify impact while controlling for scale:

$$\log(C_i) = \alpha + \beta \log(W_i) + \gamma D_i + \delta^\top Z_i + \varepsilon_i$$

Interpretation:

- $e^\gamma - 1 \approx$ percent cost increase per additional delay day

- $e^{\delta_k} - 1 \approx$ cost uplift for disruption $k$relative to "No Disruption"

This provides a managerial "delay cost curve," supporting decisions like whether to expedite or buffer inventory when delay risk increases.[2,4,1,2]

3.6 Approach(ii): forecasting delays and financial impact for the next four months

This approach is implemented as two linked forecasting problems:

1. Forecast delay dynamics (and/or delayed shipment counts), and
2. Forecast financial impact (cost uplift or expected excess cost).

3.6.1 Building the time-series target y_t

A time-series model requires a time index t(weekly or monthly). Once a timestamp is available, shipment-level uplifts are aggregated into one of the following series:

A) Mean uplift (expected financial risk):

$$y_t = \text{mean}(U_i \mid i \in t)$$

B) Extreme uplift rate (high-cost frequency):

$$y_t = \Pr\,(U_i > u \mid i \in t)$$

Optional (finance-grade tail risk): $y_t = \text{CVaR}_q(U \mid t)$, which captures tail severity beyond a threshold.[12]

3.6.2 Exogenous drivers x_t(ERP-feasible)

To improve forecast quality and interpretability, the thesis constructs exogenous drivers per period:

- Delay intensity: mean delay days or % delayed shipments
- Disruption mix: shares of each disruption category
- Volume/scale: total shipped weight or shipment count

These drivers align with the logic that risk is influenced by operational conditions, not only by past values.[4]

3.7 Forecasting models

3.7.1 ARIMA and SARIMA (classical time series baseline)

ARIMA models capture autoregressive and moving-average structure after differencing;

To forecast the delay-driven financial risk signal (e.g., weekly/monthly mean cost uplift $y_t$or the extreme-uplift rate), this thesis uses **ARIMA** and **SARIMA** as classical statistical baselines. ARIMA models the series as a combination of autoregressive (AR) and moving-average (MA) dynamics applied to a differenced version of $y_t$to achieve stationarity. In operator notation, the ARIMA(p, d, q) model is written as:

$$\phi(B)\,(1-B)^d\,y_t \;=\; c \;+\; \theta(B)\,\varepsilon_t,$$

where $B$is the backshift operator ($By_t = y_{t-1}$), $(1-B)^d$is the differencing operator of order d, $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$is the AR polynomial, $\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$is the MA polynomial, and $\varepsilon_t$is a white-noise error term.[1] Model specification follows the Box–Jenkins procedure: (i) test stationarity and apply differencing as needed; (ii) use the autocorrelation function (ACF) and partial autocorrelation function (PACF) to propose candidate orders (p, q); (iii) estimate competing models by maximum likelihood; and (iv) validate the final model using residual diagnostics, including tests for remaining autocorrelation.[2] SARIMA extends ARIMA by adding explicit **seasonal** components and seasonal differencing. The SARIMA$(p, d, q) \times (P, D, Q)_s$ model is written as:

$$\Phi(B^s)\,\phi(B)\,(1-B)^d\,(1-B^s)^D\,y_t \;=\; c \;+\; \Theta(B^s)\,\theta(B)\,\varepsilon_t,$$

where $s$is the seasonal period (e.g., $s = 12$for monthly seasonality), $(1-B^s)^D$is seasonal differencing, $\Phi(B^s) = 1 - \Phi_1 B^s - \cdots - \Phi_P B^{Ps}$is the seasonal AR polynomial, and $\Theta(B^s) = 1 + \Theta_1 B^s + \cdots + \Theta_Q B^{Qs}$is the seasonal MA polynomial.[1] In this thesis, ARIMA/SARIMA forecasts are produced using rolling-origin (walk-forward) backtesting to mimic real deployment, with accuracy reported via MAE and RMSE and later compared to alternative models in the benchmark set.[3,13, 14]

Foot Notes

1. George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel, and

Greta M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. (Hoboken, NJ: Wiley, 2015).

2. Box et al., *Time Series Analysis*, chapters on the Box–Jenkins methodology and diagnostic checking.
3. Rob J. Hyndman and George Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. (Melbourne: OTexts, 2021), sections on time-series cross-validation and forecast accuracy.

3.7.2 SARIMAX

To forecast delay-driven financial risk over the next four months, this thesis applies a **Seasonal ARIMA model with exogenous regressors (SARIMAX)** because the risk signal is driven by (i) its own time dependence (autocorrelation and possible seasonality) and (ii) operational drivers observable in ERP data (e.g., delay intensity, disruption mix, shipment volume). The model is specified as a **dynamic regression with ARIMA errors**. Let $y_t$denote the time-indexed risk metric (e.g., weekly mean cost uplift, or the weekly rate of extreme uplift events), and let $x_t \in \mathbb{R}^k$be a vector of exogenous regressors. The structural (regression) component is written as

$$y_t = c + \beta^\top x_t + u_t,$$

where $c$is an intercept, $\beta$is a coefficient vector, and $u_t$is the serially correlated disturbance. The disturbance is modeled as a seasonal ARIMA process:

$$\Phi(B^s)\,\phi(B)\,(1-B)^d\,(1-B^s)^D\,u_t \;=\; \Theta(B^s)\,\theta(B)\,\varepsilon_t,$$

where $B$is the backshift operator ($By_t = y_{t-1}$), $s$is the seasonal period (e.g., $s = 12$for monthly data), $\phi(\cdot)$and $\theta(\cdot)$are non-seasonal AR and MA polynomials, $\Phi(\cdot)$and $\Theta(\cdot)$are seasonal AR and MA polynomials, $d$and $D$are non-seasonal and seasonal differencing orders, and $\varepsilon_t$is white noise.[1] Model identification follows Box–Jenkins logic (stationarity checks and differencing, seasonal diagnostics, and selection of p, q, P, Qusing information criteria and residual tests), while exogenous regressors are included only if they are available at forecast time—otherwise they are lagged or forecasted separately to avoid look-ahead bias.[2] Forecast accuracy

is assessed using rolling-origin backtesting (walk-forward validation) and reported via MAE/RMSE; competing models are then compared statistically using the Diebold–Mariano test to support defensible conclusions about which forecasting approach performs best for ERP-style delay-risk signals.[3]

Footnotes (Chicago):

1. *statsmodels* Development Team, "SARIMAX (Seasonal ARIMA with Exogenous Regressors)," *statsmodels* documentation, accessed March 14, 2026, https://www.statsmodels.org/dev/generated/statsmodels.tsa.statespace.sarimax.SARIMAX.html.
2. George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel, and Greta M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. (Hoboken, NJ: Wiley, 2015).
3. Rob J. Hyndman and George Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. (Melbourne: OTexts, 2021), https://otexts.com/fpp3/tscv.html; Francis X. Diebold and Roberto S. Mariano, "Comparing Predictive Accuracy," *Journal of Business & Economic Statistics* 13, no. 3 (1995): 253–263.

This is particularly suitable for ERP contexts because many useful predictors (delay intensity and disruption mix) are observable. SARIMAX in state-space form is widely implemented and supports forecasting with drivers. [4,1,5]

3.7.3 Prophet

Prophet is included because supply chain risk signals often show trend shifts and changepoints during disruption waves. Prophet's decomposable structure supports interpretability and robustness for business time series, and it supports additional regressors for fair comparison with SARIMAX.[5,1,6]

To forecast the **next four months** of delay-driven financial risk, this thesis uses **Prophet** as a benchmark model that is well suited to business time series affected by regime shifts. Prophet treats the risk signal y_t(e.g., weekly/monthly mean cost uplift, or the weekly share of "high-uplift" shipments) as an additive combination of interpretable components:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

where $g(t)$is a piecewise linear (or logistic) **trend** with automatically selected **changepoints**, $s(t)$captures recurring **seasonality** using Fourier terms, $h(t)$optionally models known events/holidays, and $\epsilon_t$is the residual error.[1] In the procurement-risk context, this structure is valuable because disruption waves (e.g., congestion spikes or route diversions) frequently create abrupt shifts in the level or slope of cost risk that classical stationary models can struggle to represent without repeated re-specification; Prophet addresses this by explicitly modeling trend changes via changepoints.[1] In this thesis, Prophet is estimated in two variants: (i) a baseline model (trend + seasonality) that acts as a robust "business forecasting" comparator, and (ii) an augmented model that includes **ERP-derived regressors**—such as delay intensity and disruption-regime shares—to improve explanatory power and to enable a fair comparison with SARIMAX on a shared driver set.[1] Forecast accuracy is assessed with **rolling-origin (walk-forward) backtesting** to avoid time leakage and to replicate the operational forecasting situation, and competing models are compared using MAE/RMSE and the **Diebold–Mariano** test to determine whether performance differences are statistically meaningful.[2,3]

3.8 Count models for delay incidence: Poisson and Negative Binomial

Delays can be modeled as count processes, which is useful when forecasting:

- number of delayed shipments per week/month, and/or
- total delay-days per period.

3.8.1 Poisson regression

To forecast **delay incidence** as a count process, this thesis applies **Poisson regression** within a generalized linear modeling (GLM) framework. The dependent variable is specified as a non-negative integer count, such as (a) the **number of delayed shipments** in period t(Y_t), or (b) the **total delay-days** accumulated in period t, both of which align with how ERP exception monitoring is commonly aggregated for weekly or monthly control reporting. Under the Poisson assumption, $Y_t$follows a Poisson distribution with mean

λ_t, and the model uses a log link to ensure non-negative fitted values:

$$Y_t \sim \text{Poisson}(\lambda_t), \log(\lambda_t) = \alpha + \beta^{\top} x_t,$$

where $x_t$contains ERP-derived predictors such as delay intensity drivers, disruption regime shares (e.g., port congestion or weather exposure), and shipment volume proxies (shipment count or total weight). Model parameters are estimated by maximum likelihood, and results are interpreted through **incidence rate ratios** (IRRs), where $\exp(\beta_j)$represents the multiplicative change in the expected delay count associated with a one-unit increase in predictor $x_{j,t}$. Because operational delay counts often exhibit **overdispersion** during disruption waves (variance exceeding the mean), the Poisson model is treated as a baseline and is complemented by an overdispersion diagnostic; if overdispersion is substantial, a Negative Binomial specification is preferred for forecasting and risk quantification.[1] [2]

Footnotes:

1. P. McCullagh and J. A. Nelder, *Generalized Linear Models*, 2nd ed. (London: Chapman & Hall, 1989), chap. 6 (Poisson GLM with log link and maximum likelihood estimation).

2. A. Colin Cameron and Pravin K. Trivedi, *Regression Analysis of Count Data*, 2nd ed. (Cambridge: Cambridge University Press, 2013), chap. 3–4 (interpretation via incidence rate ratios and diagnosing overdispersion).

3.8.2 Negative Binomial regression (overdispersion)

Supply chain delays often cluster in disruption regimes, producing overdispersion (variance > mean). Negative Binomial models are used when Poisson assumptions are violated. Cameron and Trivedi provide the standard reference and practical diagnostics for this choice.[6]

To forecast **delay-related counts** in a disruption-sensitive supply chain, this thesis uses **Negative Binomial (NB) regression** because delay events in logistics rarely behave like "independent Poisson arrivals." In practice, delays cluster during port congestion waves, severe weather windows, or geopolitical route diversions, meaning the count variance typically exceeds

the mean (**overdispersion**). If a Poisson model were used under overdispersion, it would usually understate uncertainty and inflate statistical significance, which is risky for managerial decision-making.[1] Accordingly, the dependent variable is specified as a count outcome Y_t—for example, the **number of delayed shipments** in week/month t(or, alternatively, the **total delay-days** aggregated to period t). Conditional on a set of ERP-derived predictors X_t(e.g., delay intensity indicators, disruption-regime shares, and shipment volume/weight), the expected count is modeled via a log-link:

$$E(Y_t \mid X_t) = \mu_t, \log(\mu_t) = \alpha + \beta^\top X_t.$$

The NB model extends the Poisson by introducing a dispersion parameter that explicitly allows $\mathrm{Var}(Y_t \mid X_t) > E(Y_t \mid X_t)$. In the commonly used NB2 specification, the conditional variance is:

$$\mathrm{Var}(Y_t \mid X_t) = \mu_t + \kappa\mu_t^2,$$

which fits disruption-driven environments where variability increases nonlinearly as congestion or instability rises.[2] Model selection is implemented as follows: a Poisson baseline is estimated first; overdispersion is assessed using mean–variance diagnostics and residual-based tests; if overdispersion is detected, NB is adopted for forecasting because it produces more reliable parameter inference and forecast intervals for planning.[1] [3] Forecast outputs (expected delayed shipments per period) are then translated into financial exposure using the thesis' cost-impact model—i.e., predicted delay incidence increases the expected uplift in shipment cost, supporting four-month risk outlooks and mitigation planning.[4]

Footnotes:

1. A. Colin Cameron and Pravin K. Trivedi, *Regression Analysis of Count Data*, 2nd ed. (Cambridge: Cambridge University Press, 2013), 67–72, 87–95.
2. Joseph M. Hilbe, *Negative Binomial Regression*, 2nd ed. (Cambridge: Cambridge University Press, 2011), 11–18, 41–45.
3. Alan Agresti, *Foundations of Linear and Generalized Linear Models*

(Hoboken, NJ: Wiley, 2015), 196–202.
4. Sunil Chopra and ManMohan S. Sodhi, "Managing Risk to Avoid Supply-Chain Breakdown," *MIT Sloan Management Review* 46, no. 1 (Fall 2004): 53–61.

3.9 From predicted delays to predicted financial loss

Forecasts only create value if they can be expressed in money and linked to an action. In this thesis, the "financial loss" we care about is the avoidable freight cost escalation caused by delay days and disruption regimes, over and above the baseline cost explained by shipment scale (weight). To translate predictive outputs into finance-ready numbers, the methodology uses two core conversion methods, and adds a third (new) method that produces a full loss distribution (expected + worst-case exposure) that procurement and finance teams can directly use for budgeting and risk appetite decisions.[1]

3.9.1 Regression-based conversion

The cost-impact model is estimated at shipment level in log form:

$$\log(C_i) = \alpha + \beta \log(W_i) + \gamma D_i + \delta^\top Z_i + \varepsilon_i$$

where C_i is shipping cost, W_i is shipment weight, D_i is delay days, and Z_i represents disruption regime indicators.

Step 1 — convert delay forecast to expected % cost change. The coefficient γ is a semi-elasticity. If the expected delay changes by ΔD, the expected proportional cost change is:

$$\%\Delta C \approx \exp(\gamma \Delta D) - 1$$

This is the most direct conversion for procurement because it gives an immediately interpretable rule: *"Each additional expected day of delay increases expected transport cost by X% (holding shipment scale and regime constant)."* It supports practical decisions like expedite vs buffer and the level at which expediting must be approved. [2]

Step 2 — express this in currency for a specific shipment. If you have a baseline cost estimate $\hat{C}_i^{base}$ (based on weight and normal conditions), then expected additional cost is:

$$\widehat{Loss}_i \approx \hat{C}_i^{base} \cdot (\exp(\gamma \Delta D) - 1)$$

Important technical detail (so your totals are not biased). Because the model is fitted on $\log(C)$, converting predicted log-cost back to cost levels can introduce retransformation bias. A standard practical correction is the Duan smearing estimator, which adjusts for the fact that $E[\exp(\varepsilon)] \neq \exp(E[\varepsilon])$.[3] This matters when you aggregate predicted costs into monthly exposure numbers.

What your supervisor will like about this method: it is transparent, explainable, and auditable—every number can be traced to a coefficient and a baseline.

3.9.2 Period exposure forecast

Procurement and finance typically manage exposure by week or month, not shipment by shipment. The second conversion method therefore estimates expected excess freight spend per period.

Let y_t be a time-series risk measure such as mean uplift in period t:

$$y_t = \text{mean}(U_i \mid i \in t)$$

Let $\hat{y}_t$ be its forecast from ARIMA/SARIMA/SARIMAX/Prophet. Let $\widehat{BaselineCost}_t$ be expected baseline freight spend (what freight would cost under normal conditions given expected volume/weight). Then:

$$\widehat{Exposure}_t \approx \hat{y}_t \cdot \widehat{BaselineCost}_t$$

Four-month exposure (monthly aggregation) is then:

$$\widehat{Exposure}_{t:t+4} = \sum_{\tau=t+1}^{t+4} \widehat{Exposure}_\tau$$

How to estimate $\widehat{BaselineCost}_t$ in practice (ERP reality):

- If you can forecast total weight/volume W_t^{tot}, you can derive baseline using the baseline cost function.
- If volume forecasting is not feasible, you can use a seasonal baseline (last-year same month baseline cost) and scale it by expected demand or

shipment count.[4]

Why this is finance-ready: it yields the sentence that managers need: "Expected additional freight cost over the next four months due to delay/disruption risk is €X (with an uncertainty range)."

3.9.3 New method: loss-distribution forecasting using a frequency–severity model

The two methods above give good expected exposure. But disruption risk is often about tail events (rare periods with very high uplift). Therefore, the thesis introduces a third method drawn from actuarial/operational risk practice: model total loss as a compound (frequency–severity) process:

$$L_t = \sum_{j=1}^{N_t} S_{t,j}$$

- N_t: number of "loss events" in period t(e.g., delayed shipments, or shipments in high-cost regimes)
- $S_{t,j}$: severity of each event (excess cost per shipment)

This is the classic collective risk / compound loss framework used to quantify expected and tail loss distributions.[5] [6]

(a) Frequency model N_t: how many delay-affected shipments next month?

We forecast $N_t$using your planned models:

- Poisson regression (baseline),
- Negative Binomial regression (if overdispersion exists), with operational drivers x_t(delay intensity, disruption shares, volume).

Poisson:

$$N_t \sim \text{Poisson}(\lambda_t), \log(\lambda_t) = \alpha + \beta^\top x_t$$

Negative Binomial (overdispersion):

$$E[N_t \mid x_t] = \mu_t, \log(\mu_t) = \alpha + \beta^\top x_t, \text{Var}(N_t \mid x_t) = \mu_t + \kappa\mu_t^2$$

This matches standard count modeling practice for clustered operational events.[7]

(b) Severity model $S_{t,j}$: how expensive is each delay event?

Define severity as excess cost:

$$S_i = C_i - \hat{C}_i^{base} \text{or} S_i = \hat{C}_i^{base} \cdot U_i$$

Then model severity conditional on weight and regime. Two practical options:

Option 1: Lognormal/Gamma severity regression (mean severity).

$$\log(S_i) = a + b\log(W_i) + cD_i + d^\top Z_i + \eta_i$$

This aligns with heavy-tailed cost behavior and produces interpretable marginal effects.

Option 2 (recommended for disruption risk): Quantile regression for severity (tail-aware).

Instead of modeling the mean, quantile regression models the conditional quantiles $Q_\tau(S \mid X)$(e.g., 90th percentile severity). This is useful because disruption risk is usually about the high tail. Quantile regression is a standard tool for this purpose.[8]

(c) Loss simulation: produce expected loss *and* tail loss (VaR/CVaR)

Once frequency and severity are modeled, we generate a loss distribution for each forecast month:

1. simulate $N_t$from Poisson/NB forecast distribution
2. simulate $S_{t,1}, \ldots, S_{t,N_t}$from the fitted severity model
3. compute $L_t = \sum S_{t,j}$
4. repeat many times to obtain a distribution of monthly loss

Then report:

- $E[L_t]$: expected monthly financial loss
- $\text{VaR}_q(L_t)$: worst-case quantile
- $\text{CVaR}_q(L_t)$: expected loss in the worst tail

CVaR/Expected Shortfall is widely used in risk management because it captures tail severity beyond a single quantile. [9] [10]

Why this new method is valuable:

It gives procurement and finance a robust statement like:

"Next month's expected excess freight cost is €X, and in the worst 10% of

scenarios it exceeds €Y."

This is far more aligned with real risk governance than reporting only point forecasts.

3.10 Model evaluation: RMSE, MAE, and Diebold–Mariano (DM) test

Because this thesis compares multiple models (ARIMA/SARIMA/SARIMAX/Prophet plus Poisson/NB), the evaluation must be time-consistent and statistically defensible.

3.10.1 Rolling-origin backtesting

Forecast evaluation uses rolling-origin backtesting:

- train on data up to time t
- forecast $t+1, \ldots, t+h$
- move forward and repeat

This preserves time order and mimics operational deployment. It is the recommended approach in applied forecasting.[11]

3.10.2 Accuracy metrics: MAE and RMSE

For each model and horizon, report:

$$MAE = \frac{1}{n}\sum_{t} | y_t - \hat{y}_t | \quad RMSE = \sqrt{\frac{1}{n}\sum_{t}(y_t - \hat{y}_t)^2}$$

- MAE: easier to explain and robust to outliers
- RMSE: penalizes large errors, which matters when spikes are expensive

Forecast accuracy literature recommends reporting multiple metrics because each emphasizes different decision concerns.[12] [13]

3.10.3 DM test (statistical comparison) + HAC variance

To test whether forecast accuracy differences are statistically significant, apply the DM test. Let $e_{1,t}$and $e_{2,t}$be errors and define loss differential:

$$d_t = L(e_{1,t}) - L(e_{2,t})$$

Test $H_0: E[d_t] = 0$. The variance of $\bar{d}$should allow autocorrelation in multi-step horizons; a HAC estimator (often Newey–West) is standard.[14] [15]

Because samples can be limited (especially with monthly ERP series), apply the Harvey–Leybourne–Newbold adjustment for better small-sample

performance.[16]

3.11 Approach (iii): linking predictive insights to mitigation strategy (turning forecasts into cost reduction)

The value of forecasting is realized only when it changes decisions and reduces consequence severity. Resilience research emphasizes that capability is measured in *recovery speed and consequence reduction*, not in having a model.[17]

3.11.1 Governance-ready triggers (simple and auditable)

The thesis proposes practical triggers that an ERP analytics dashboard can operationalize:

- Expected-risk trigger: if forecast mean uplift $\hat{y}_t$exceeds a threshold → escalation
- Tail-risk trigger: if forecast CVaR (from the new frequency–severity method) exceeds budget tolerance → contingency actions activated
- Regime trigger: if forecast share of high-cost regimes rises (e.g., congestion/diversion) → pre-emptive capacity/routing actions

This aligns with ERM guidance: define thresholds, assign ownership, and activate treatment plans.[18] [19]

3.11.2 Action playbooks linked to drivers

(1) High predicted congestion / route diversion (regime shift)

- reserve capacity earlier,
- reroute planning and alternative carriers,
- contract flexibility and pre-approved exception handling,
- targeted buffers for critical inbound flows.[1] [17] [20]

(2) High predicted delay incidence (frequency spike)

- receiving capacity planning,
- PO release timing adjustments,
- split shipments and scheduling discipline,
- supplier escalation and reliability governance.[1] [20]

(3) High predicted cost uplift (severity spike)

- approval gates for premium freight,
- audit of accessorial charges and carrier compliance,

- targeted supplier/lane reviews and chargebacks where feasible.[21]

This is consistent with mitigation economics: mitigation is justified when it reduces expected loss and/or tail loss more than it costs.[22]

Footnotes:

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